HEIRSTONE CONSULTING

DIGITAL INFRASTRUCTURE REAL ESTATE

Borrowed ground

Towers, fibre and the land beneath Africa's digital networks, and why the ground lease and the wayleave are their weak links



THE REPORT IN SIX FIGURES

89%

of IHS Towers' sites stand on leased land, with 11.4 years left on average; SBA owns or controls 71% of its sites' land for more than 20 years

US$72.5k to 325k

value per site in African tower deals, from MTN South Africa (2022) to Eaton Towers (2019-20)

4x

power cost per tower site against ground rent: about US$12,600 to 14,300 a year against US$3,100 to 3,500

7 to 28%

Ogun State's one-off fibre right-of-way charge as a share of Nigeria's average Project BRIDGE build budget per km, on ALTON and TechCabal figures (2025)

11 to 15x

revenue per km earned by Dark Fibre Africa (2025-26) against Kenya Power's dark fibre on its grid (2022-23)

116m

people per cable landing site in Nigeria, against 13m in Egypt, which spreads 24 systems, some still planned, over nine sites, and 8m in South Africa

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THE ARGUMENT

Africa's towers and fibre are valued as long-life infrastructure, yet they stand on short land rights paid in local currency and negotiated plot by plot or state by state. Much of the value still to be created in digital real estate sits in making those rights longer, indexed and easier to aggregate, and in bringing power to the plot.

ABOUT THIS REPORT

This report looks at African towers and fibre as real estate: the plots, rooftops, streets, corridors and beaches that digital networks occupy, who holds the rights to them and on what terms. It is written for telecom operators and tower companies, infrastructure investors, pension funds and development finance institutions, landowners, road, rail and power authorities, regulators and governments in North and Sub-Saharan Africa.

It draws only on public sources current in 2026: company filings and results, court judgments, regulator and ministry publications, cable registries and the trade press. Figures we derive, such as value per site, cost per kilometre or people per landing site, are labelled Heirstone analysis and show their inputs.

Local-currency figures are given with a US$ equivalent and the exchange-rate date. Heirstone publishes a separate report on data centres, which are not covered here.

SUMMARY

Africa's towers and fibre are financed as long-life infrastructure, on land held for a fraction of that life

Investors buy African tower and fibre businesses for their long, largely hard-currency contracts with mobile operators. The land rights under them run for a fraction of that time and are paid in local currency, agreed one landowner, state or municipality at a time. That mismatch matters more than the level of rent.

1 **The tower sits on a short lease.**

About 89% of IHS Towers' sites stand on ground leases with an average of 11.4 years left, and a third expire before the end of 2030. SBA Communications, a US-listed owner, holds 71% of its sites' land outright or for over 20 years, with 35 years of lease life left including renewal options. Applied to IHS's African towers, the group-wide share implies about 25,500 ground leases.

2 **Buyers pay for tenants and contracts. The ground explains little of the price.**

African tower deals have valued sites at between about US$72,500 (MTN South Africa, 2022) and US$325,000 (Eaton Towers, 2019-20). Capitalised at 8 to 12%, a typical ground rent would account for 36 to 60% of the MTN South Africa price per site but only 8 to 13% of Eaton's.

3 **Power is the real occupancy cost.**

A tower site's power costs about US$12,600 to US$14,300 a year, four times its ground rent of US$3,100 to US$3,500. Power was 47.8% of IHS's cost of sales in 2025, with grid supply across Helios Towers' markets averaging 16 to 18 hours a day.

4 **In fibre, the permit can cost up to a quarter of the build.**

On TechCabal's 2025 compilation, Nigerian states charged from nothing to N9,477 per metre for right of way. At that rate Ogun's one-off charge equals about 28% of Project BRIDGE's average budget of US$22,200 per km; ALTON, the operators' association, puts Ogun at over N2,500, or at least 7%. The N145 federal benchmark fell from about US$190 to US$94 per km between mid-2023 and mid-2025.

5 **City ducts earn far more than long corridors.**

Dark Fibre Africa earned 11 to 15 times more revenue per kilometre in 2025-26 than Kenya Power's dark fibre on its grid did in 2022-23 (about US$770 per km a year). Vodacom's R12.64bn (US$712m) for 30% of Maziv implies US$813 to US$1,160 per home passed.

6 **Landing sites are few and concentrated.**

Nigeria's eight international cables land at two sites, one for every 116 million people, against one for every 13 million in Egypt, where 24 systems, some still planned, land at nine sites. Dakar, Accra and Abidjan each depend on a single landing site.

THE MODEL IS TURNING

MTN sold towers to American Tower and IHS between 2011 and 2022. In August 2026 IHS shareholders approved MTN's offer for the 75% of IHS it did not already own. IHS put the enterprise value of the sale at US$6.2bn.

Paradigm Tower Ventures, backed by British International Investment and Proparco, bought IHS Rwanda in 2025. Paradigm Infrastructure is SBA's minority partner in Tanzania.

25,500

ground leases implied under IHS's African towers (Heirstone estimate from IHS 20-F)

3,600

of them due for renewal within two years

US$22,200

average budget per route-km for Nigeria's 90,000 km Project BRIDGE

SUMMARY, CONTINUED

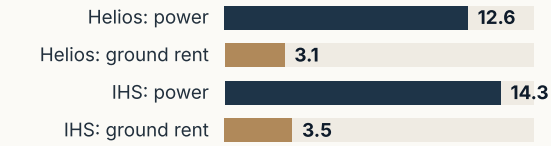
Land is the cheapest input in African digital infrastructure and the one held on the weakest terms

EXHIBIT 1

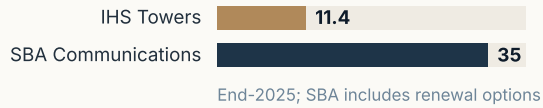
The ground costs little. The terms on which it is held carry the risk

Selected measures for tower sites (Helios FY2025; IHS H1 2026 annualised) and fibre rights of way in Nigeria (2025)

Tower site costs, US$ thousand per site a year

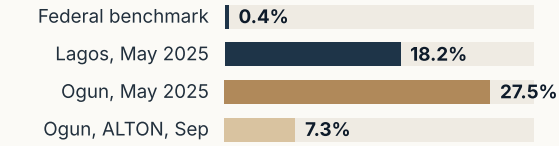


Average remaining ground-lease life, years



End-2025; SBA includes renewal options

Nigeria: one-off right of way as % of build cost per km



Nigeria: N145 per metre benchmark in US$ per km



Source: Helios Towers FY2025 results (12 Mar 2026); IHS Holding 2Q26 financial statements and Form 20-F FY2025; SBA Communications Form 10-K FY2025; TechCabal state right-of-way table (26 May 2025) and TechCabal citing ALTON (12 Sep 2025: Ogun over N2,500 per metre, a lower bound); Federal Ministry of Communications, Innovation and Digital Economy, Project BRIDGE (US$2bn for 90,000 km); US Treasury reporting rates (NGN 765, 1,550 and 1,326.78 per US$ at 30 Jun 2023, 30 Jun 2025 and 30 Sep 2026). **Heirstone analysis.** Per-site costs divide reported power and lease payments by sites.

The panels above carry the argument of this report. On a tower site, ground rent is the smallest of the costs that matter: about a quarter of the power bill and around 5% of revenue. The lease is short, at 11.4 years remaining on average across IHS's portfolio against 35 years at SBA. In fibre, on published Nigerian state rates, the price of the right to dig ranges from nothing to more than a quarter of the build cost, depending on which state the route crosses, and the federal benchmark lost half its dollar value between mid-2023 and mid-2025.

Rent per site and charge per metre are small numbers, and both erode with the currency. Term, indexation, the number of counterparties, certainty of consent, the right to add tenants and access to power decide what a tower or a fibre route is worth to its owner and its lenders. Chapter 7 turns these six into a test that any landowner, authority or investor can apply.

5.4%

Helios Towers' lease payments as a share of revenue in 2025, against 21.7% for power

17.7x

Maziv enterprise value over operating profit implied by Vodacom's entry

WHAT FOLLOWS FOR EACH AUDIENCE

Governments, regulators and land agencies

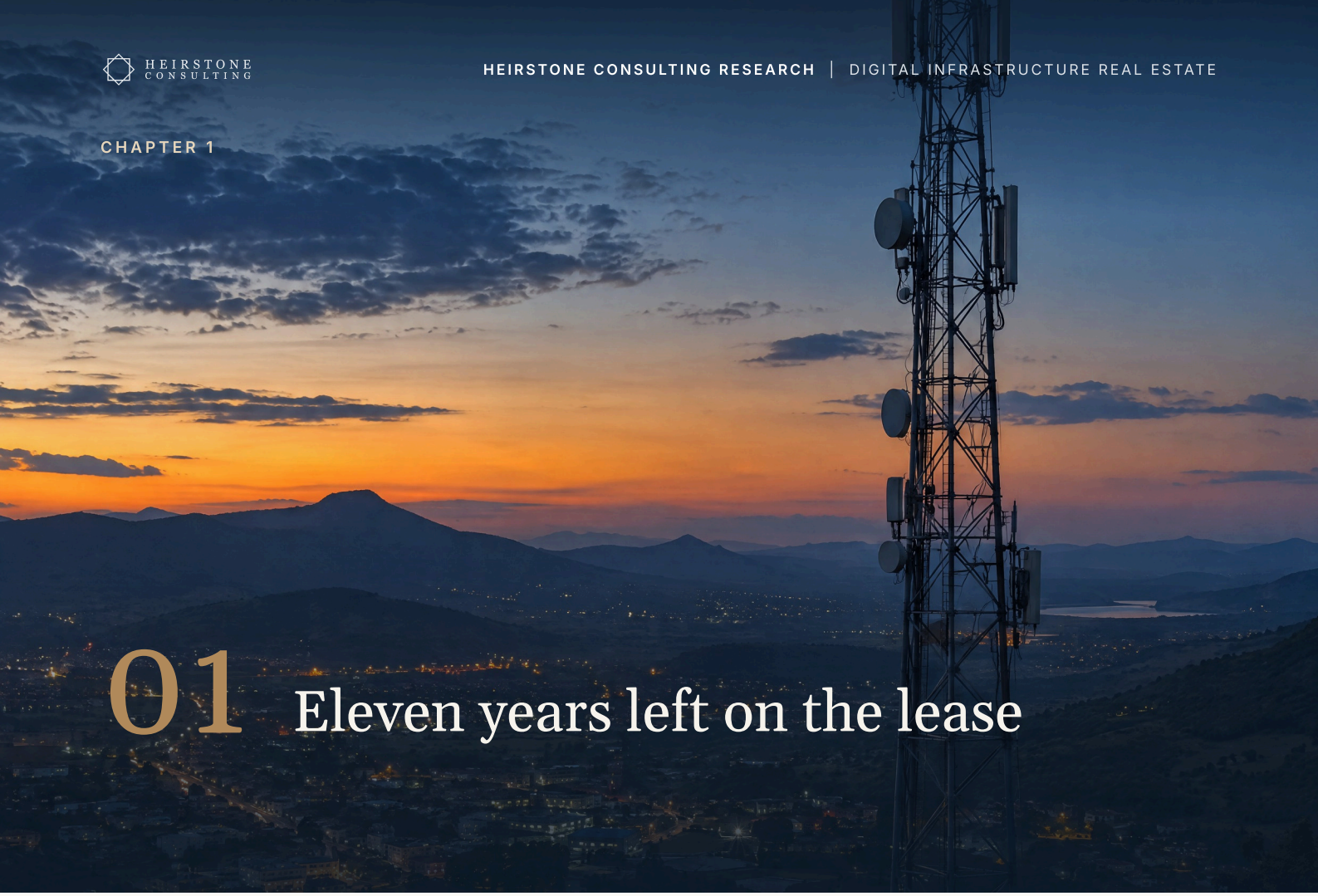
Publish one national right-of-way tariff, index it and collect it through one window. Reserve landing zones and duct routes in coastal, road and rail plans. Offer state land for tower sites on standard 20-year leases.

Landowners and corridor owners

Sell term, indexation and power before asking for more rent. Road, rail and power authorities should lease ducts on the urban approaches, where fibre earns, and price long-haul corridors as volume business.

Operators, investors and DFIs

Underwrite the lease-expiry profile and the currency of the land rights alongside tenancy. Lengthened and aggregated, the ground leases under African towers could become a local-currency asset for pension funds.



CHAPTER 1

01 Eleven years left on the lease

Africa's independent tower companies own the steel. Most of them rent the ground it stands on, under tens of thousands of separate ground leases that are short by the standards of the investors who buy the towers.

IHS Towers, which describes itself as Africa's market leader by tower count, had 28,662 towers on the continent at the end of 2025. About 89% of its sites operate under ground leases on land it does not own. The leases typically run for ten to fifteen years and are paid monthly, quarterly or several years in advance. At the end of 2025 their average remaining life was 11.4 years. About a third expire before the end of 2030 and 14% were due for renewal within 24 months.

Applied to the 28,637 towers in IHS's continuing African business at mid-2026, the group-wide share implies about 25,500 ground leases, of which about 3,600 come up for renewal within two years. Each is a separate agreement with a landowner, paid in local currency.

SBA Communications, a US-listed tower owner with sites in the United States and international markets, shows what a mature portfolio looks like. About 71% of its towers stand on land it owns or controls for more than 20 years, and its ground leases have 35 years left on average, including renewal options.

28,662

IHS towers in Africa at the end of 2025; IHS describes itself as the market leader by tower count

11.4 yrs

average remaining life of IHS ground leases, against 35 years at SBA

TENURE

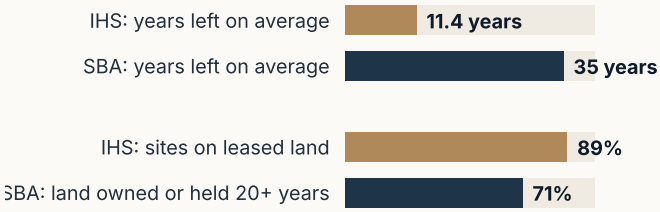
Tower valuations rarely show the lease. It caps both the tenancy and the margin they do show.

EXHIBIT 2

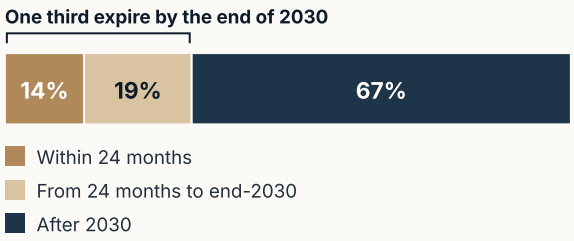
IHS ground leases have about a third of the remaining life of SBA's

Ground-lease tenure of IHS Towers (group, including Latin America) and SBA Communications (United States and international), 31 December 2025

Remaining ground-lease life and land control, end-2025



IHS ground leases by expiry, share of leases



Source: IHS Holding, Form 20-F FY2025 (89% of sites on ground leases on land it does not own; 11.4 years average remaining life; about 33% expire before end-2030; 14% due within 24 months); SBA Communications, Form 10-K FY2025 (71% on land owned or controlled for more than 20 years; 35 years including renewal options). **Heirstone analysis.** The 24-months-to-2030 band is the difference between the two IHS figures.

Tower businesses are valued on tenants per tower, EBITDA margins and contract escalators. The ground lease sits beneath all three. A tower whose lease is not renewed has to be moved or switched off, whatever its tenancy. Renewals are routine, which is why tenure attracts little attention, but each one is a negotiation with a landowner and each reopens the price.

As an illustration of what term is worth, a level income stream that lasts 11.4 years is worth 63 to 74% of the same stream lasting 35 years, at discount rates of 8 to 12%. Renewal options narrow that gap. They are worth most when they are written into the lease at the start, at a price agreed in advance.

IHS paid US$66.0m in lease cash, principal and interest, in the first half of 2026, up from US$53.1m a year earlier, and carried lease liabilities of US$379.9m at the end of June 2026. Both figures include its Latin American business, sold in August 2026.

The same exposure applies wherever towers stand on rented plots. We found no public disclosure of ground-lease terms from Helios Towers, American Tower's African business or the mobile operators that still own their own sites.

> *About 3,600 IHS ground leases come up for renewal within two years.*

14%

IHS ground leases due for renewal within 24 months of end-2025

10 to 15

years: typical term of a new IHS ground lease

US$379.9m

IHS lease liabilities at 30 June 2026, including Latin America

63 to 74%

value of an 11.4-year income stream relative to a 35-year one at 8 to 12% (Heirstone illustration)

METHOD

How we estimate 25,500 ground leases

IHS reports its lease shares for the whole group at the end of 2025. We apply them to the 28,637 towers in its continuing African business at mid-2026, after the sale of Latin America: 89% on ground leases gives 25,487 leases; 14% due within 24 months gives about 3,570; 33% expiring before 2030 gives about 8,400. The estimate assumes one ground lease per leased site and that the African portfolio matches the group average. It is an order of magnitude, not a count.

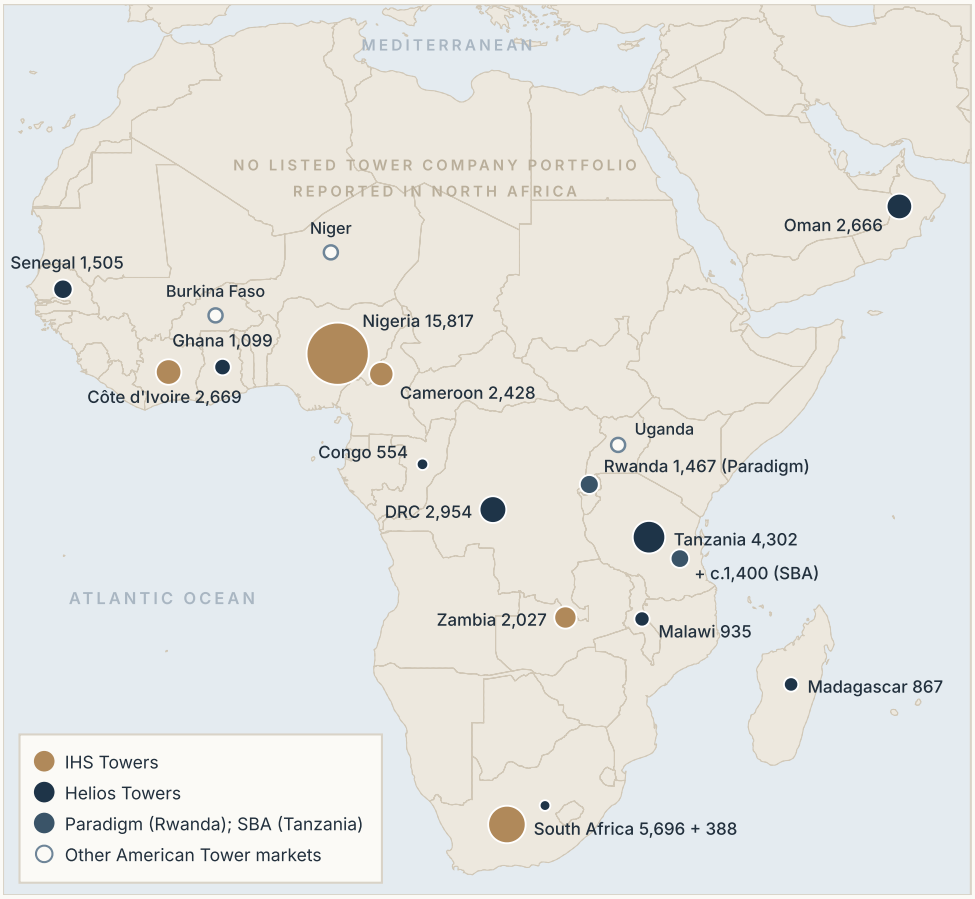
WHO OWNS THE STEEL

A handful of listed owners hold the independent towers, none of them in North Africa

EXHIBIT 3

Listed tower companies report about 44,000 African sites by country, none of them in North Africa

Sites by owner and country, 30 June 2026 (Rwanda at sale, October 2025; SBA Tanzania at acquisition, 2021); circle area scales with sites



Source: IHS Holding 2Q26 earnings presentation; Helios Towers H1 2026 results; SBA Communications 8-K (2 Jun 2021); American Tower 10-K FY2011 and FY2019 (rings show American Tower markets named in those filings beyond those already plotted: Ghana, South Africa, Uganda, Burkina Faso and Niger; country counts are not disclosed). Circles at approximate country centres. **Heirstone analysis.**

28,637

IHS towers in five African markets at mid-2026, after exits from Kuwait, Rwanda and Latin America

15,270

Helios Towers sites in eight African markets and Oman at mid-2026

18,370

American Tower sites in Africa at end-2019; it now reports Africa with Asia-Pacific (27,857 sites at end-2025)

5,800

sites planned under IHS Egypt with Egypt Digital Company for Investment, about 50 independent sites per million Egyptians: the scale an independent tower market in Egypt could reach

The independent tower model reached Africa in 2011, when American Tower acquired and built about 3,230 sites in Ghana and South Africa and agreed a joint venture with MTN in Uganda. In 2014 IHS agreed to buy the remaining 4,696 of MTN Nigeria's towers, a deal that closed in 2015. Helios Towers bought portfolios from Free Senegal, Airtel (Madagascar and Malawi) and Omantel in 2021 and 2022, American Tower closed its purchase of Eaton Towers in January 2020, and IHS bought MTN South Africa's 5,701 towers in 2022.

The owners are now concentrating. IHS sold its Kuwait business in December 2024, Rwanda in October 2025 and Latin America in August 2026, and reports five markets: Nigeria, Cameroon, Côte d'Ivoire, South Africa and Zambia. None of the listed tower companies we reviewed reports an operating portfolio in North Africa.

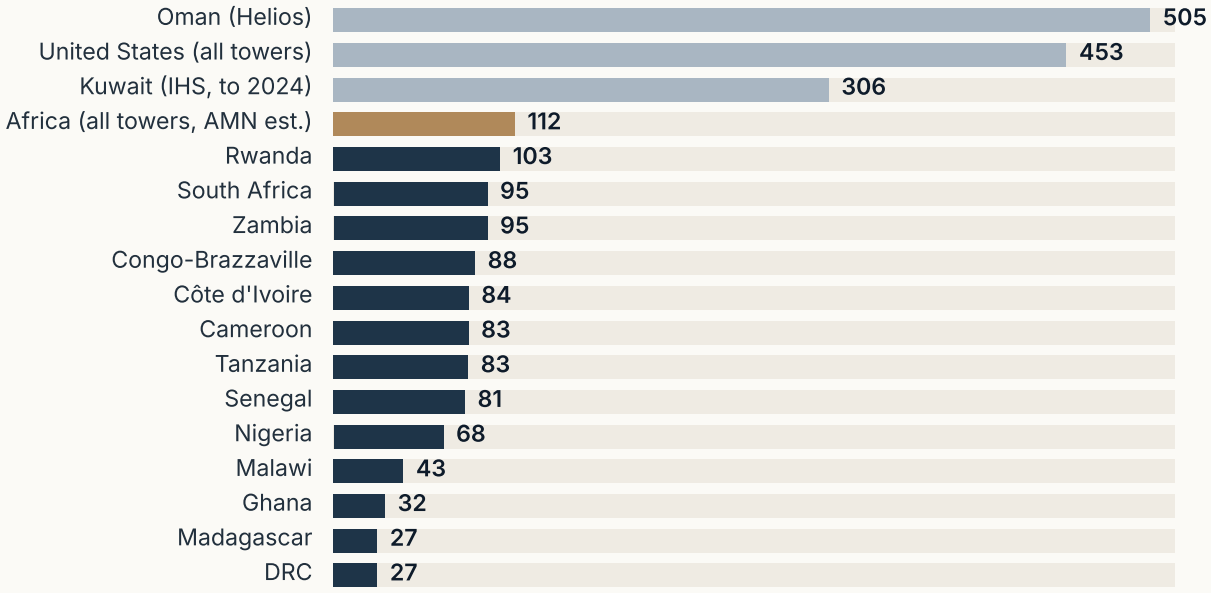
DENSITY

Listed owners hold 27 sites per million people in DRC and Madagascar and about 100 in Rwanda

EXHIBIT 4

Africa has about a quarter of the US tower density on one company's estimate; listed owners hold 27 to 103 sites per million people

Sites held by listed tower companies per million people, by country (lower bound: excludes American Tower and operator-owned sites); all-tower estimates for Africa and the US in gold and grey; 2024 population



Source: IHS 2Q26 presentation; Helios H1 2026 results; SBA 8-K (2021); IHS 20-F FY2023 (Kuwait); CTIA (154,000 US towers); Africa Mobile Networks website (170,000 African towers, a single company estimate); World Bank WDI population 2024. **Heirstone analysis.** Tanzania adds SBA's 2021 acquisition count to Helios; South Africa adds Helios to IHS.

The country figures in Exhibit 4 count only listed tower companies, so they are a floor: operator-owned sites and American Tower's African towers, no longer disclosed by country, are excluded. Even so, Rwanda, South Africa and Zambia have three to four times the listed-owner density of DRC, Madagascar and Ghana.

For a continental figure we rely on a single company estimate. Africa Mobile Networks, which builds rural sites, puts Africa's tower stock at 170,000 and says the continent needs at least 60,000 more. That is about 112 towers per million people, against about 453 in the United States on CTIA's count of 154,000. Helios's Omani portfolio, at 505 sites per million, shows how dense a Gulf market can be.

If the additional towers were sited the way IHS's are, with nine in ten on leased plots, they would add more than 50,000 ground leases, on terms set by the landowners and land agencies that grant them.

In Egypt, IHS planned to enter through IHS Egypt, owned 80% by IHS and 20% by Egypt Digital Company for Investment, with a commitment to cover 5,800 sites over three years, about 50 independent sites per million Egyptians. IHS does not list Egypt among its 2026 operating markets, so the plan remains a guide to the scale an independent tower market in Egypt could reach.

112

towers per million people in Africa on Africa Mobile Networks' estimate, against about 453 in the US

60,000

more towers Africa needs, according to Africa Mobile Networks

158m

people covered by Helios Towers' sites in 2025; its target is close to 200 million by 2030

CHAPTER 2

02

Tenants set the price

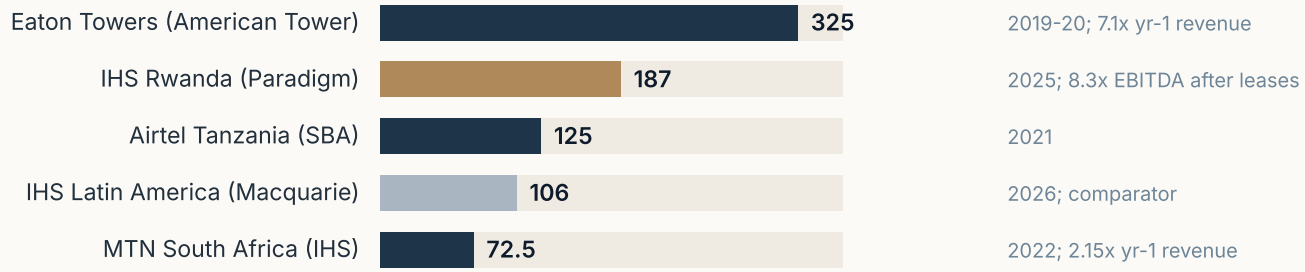
African tower portfolios have changed hands at between about US$72,500 and US$325,000 a site. The spread follows the number of tenants on each tower, the currency of the contracts and whether power is sold with the space. The land under the towers explains little of it.

Eaton Towers cost American Tower about US$1.85bn including debt, about US$325,000 a site and 7.1 times expected first-year property revenue. IHS paid ZAR 6.4bn (US$413m as reported at closing) for MTN South Africa's 5,701 towers in 2022, about US$72,500 a tower and 2.15 times expected first-year revenue of US$192m, which included power services on about 13,000 MTN sites.

EXHIBIT 5

African tower deals have valued sites at between US$72,500 and US$325,000

Transaction value per site, US$ thousand, with the multiple where disclosed or derivable



Source: American Tower press releases (30 May 2019; 2 Jan 2020); Developing Telecoms (Oct 2025); SBA 8-K (2 Jun 2021); IHS 1Q26 release; Capacity (Jun 2022). **Heirstone analysis.** Price divided by sites at closing. Eaton price includes assumed debt; Latin America is enterprise value.

Paradigm Tower Ventures, a consortium of Convergence Partners, British International Investment and Proparco, paid US$274.5m for IHS Rwanda's 1,467 towers in 2025: about US$187,000 a site, reported as 8.3 times EBITDA after leases, for a portfolio with 2.07 tenants per tower. SBA agreed in 2021 to pay about US$175m for roughly 1,400 Airtel towers in Tanzania, about US$125,000 each, with Airtel leasing space back and committing to five years of build-to-suit sites. IHS booked a gain of US$177.7m on the Rwanda sale.

7.1x

Eaton price over expected first-year property revenue of US$260m

2.15x

MTN South Africa price over expected first-year revenue

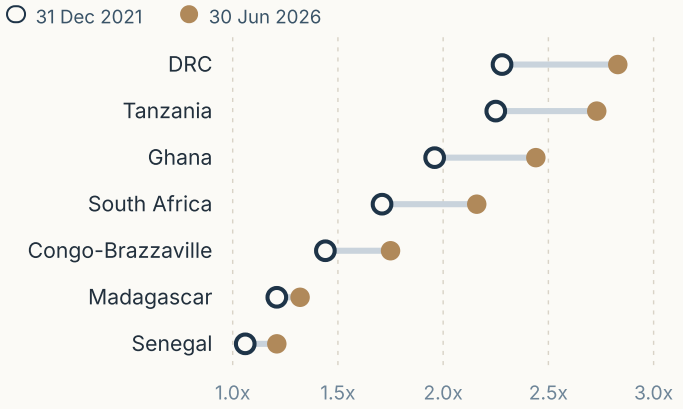
COLOCATION

A third tenant takes a new Helios site's return on capital from 12% to 34%

EXHIBIT 6

Tenancy rose in all seven Helios markets with 2021 data; Senegal and Madagascar barely moved

Tenancies per site, Helios Towers, 31 December 2021 (ring) and 30 June 2026 (dot)

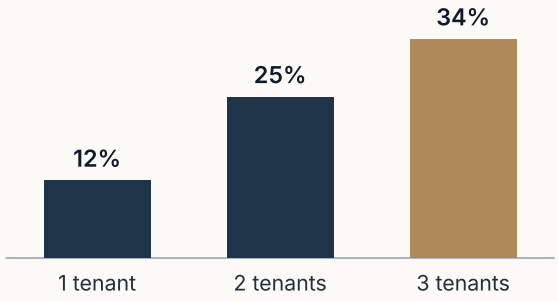


Source: Helios Towers Annual Report 2021; Helios Towers H1 2026 results (30 Jul 2026).

EXHIBIT 7

Each added tenant lifts the return on a new site

Helios day-one build-to-suit site return on invested capital, by number of tenants



Source: Helios Towers Annual Report 2025.

Helios Towers puts the day-one return on a new build-to-suit site at 12%, rising to 25% with a second tenant and 34% with a third. Its tenancy ratio rose from 2.17 to 2.26 between the end of 2025 and mid-2026, its return on invested capital was 13.5% in 2025, and it targets a ratio above 2.5 and a return of 15 to 20% by 2030.

The gains are uneven. DRC rose from 2.28 to 2.83 tenants per site between 2021 and mid-2026, Tanzania from 2.25 to 2.73 and Ghana from 1.96 to 2.44. Senegal moved from 1.06 to 1.21 and Madagascar from 1.21 to 1.32; both portfolios were bought from a single operator in 2021.

In Nigeria the direction has reversed. IHS's Nigerian tenants fell 9.1% to 23,187 in the year to mid-2026 after MTN agreed to vacate about 1,050 sites and T2, formerly 9mobile, ended 2,576 tenancies. Lease amendments, where an existing tenant adds equipment, rose 15.4% to 40,541 over the same year.

For landowners there are two consequences. Growth on an existing plot now comes from more equipment on the same tower, and some plots will be handed back as operators consolidate. We found no published evidence on whether African ground leases share colocation income with the landowner. At Helios, lease payments equal about US$1,450 per tenancy a year, against revenue of about US$26,700 per tenancy.

2.83x

tenancies per site in DRC at mid-2026, the highest in Helios's portfolio

1,050

IHS sites MTN Nigeria agreed to vacate under contract renewals

40,541

IHS Nigeria lease amendments at mid-2026, up 15.4% in a year as tenants fell 9.1%

TRANSACTIONS

MTN sold its towers to specialists between 2011 and 2022. In 2026 it is buying the largest of them.

EXHIBIT 8

Fifteen years of African tower deals end with the largest seller buying back in

Selected tower transactions in Africa, with Gulf and Latin American comparators; local-currency prices with the US$ figure reported at closing

YEAR	BUYER	SELLER	MARKET	SITES	PRICE	US$K PER SITE	NOTE
2011	**American Tower**	Various; MTN JV in Uganda	Ghana, S. Africa, Uganda	3,230	n/d		Sites acquired and built in Ghana and SA
2014	**IHS Towers**	MTN Nigeria	Nigeria	4,696	n/d		Remaining MTN Nigeria towers; closed 2015
2017-20	**IHS (70%), Zain (30%)**	Zain Kuwait	Kuwait	1,499	n/d		Sold by IHS Dec 2024
2019-20	**American Tower**	Eaton Towers holders	Five markets	c.5,700	US$1.85bn incl. debt	325	7.1x year-1 property revenue
2020	**American Tower**	MTN Group	Ghana, Uganda	JV stakes	US$523m		MTN exits its tower joint ventures
2021	**SBA, Paradigm**	Airtel Tanzania	Tanzania	c.1,400	US$175m	125	Leaseback; five years of build-to-suit
2021	**Helios Towers**	Free Senegal	Senegal	1,207	n/d		400 build-to-suit sites committed
2021-22	**Helios Towers**	Airtel Africa	Madagascar, Malawi	490; 723	n/d		Gabon (459) announced; Chad lapsed
2022	**Helios Towers**	Omantel	Oman	2,519	n/d		Gulf comparator
2022	**IHS (70%), B-BBEE (30%)**	MTN South Africa	South Africa	5,701	ZAR 6.4bn (US$413m)	72.5	2.15x year-1 revenue incl. power
2025	**Paradigm Tower Ventures**	IHS Towers	Rwanda	1,467	US$274.5m	187	8.3x EBITDA after leases
2026	**Macquarie AM**	IHS Towers	Brazil, Colombia	c.9,000	US$952m EV	106	Comparator
2026	**MTN Group**	IHS holders (75%)	Five IHS markets	c.37,600	US$6.2bn EV		Enterprise value per IHS; approved Aug 2026

Source: American Tower 10-K FY2011, press releases (2019, 2020); IHS Holding 20-F FY2021, FY2023, FY2025, 1Q26 and 2Q26 releases; IHS Towers, Our history (2026); SBA 8-K (2021); Helios Towers Annual Reports 2021 and 2022; Capacity (Jun 2022); Developing Telecoms (Oct 2025). n/d: not disclosed; n/v: not verified. **Heirstone analysis** for value per site.

MTN was the anchor seller of the African tower market. It sold its stakes in its Ghana and Uganda joint ventures with American Tower for about US$523m in 2020; IHS bought the last 4,696 of its Nigerian towers in a deal agreed in 2014 and its South African portfolio in 2022. In February 2026 MTN proposed to buy the 75% of IHS it did not own, and IHS shareholders approved the deal in August 2026. IHS gave the sale an enterprise value of US$6.2bn.

Development finance is moving in as IHS sells smaller markets. Paradigm Tower Ventures, backed by British International Investment and Proparco, bought IHS Rwanda, and Paradigm Infrastructure is SBA's partner in Tanzania.

For landowners the identity of the tenant matters. An operator that owns its towers again decides which plots its network still needs: MTN Nigeria's renegotiation with IHS put about 2,500 sites in discussion in 2023 and ended with an agreement to vacate about 1,050.

US$6.2bn

enterprise value of IHS in MTN's offer, as stated by IHS; approved by IHS shareholders in August 2026

5

IHS operating markets in 2026: Nigeria, Cameroon, Côte d'Ivoire, South Africa and Zambia

CHAPTER 3

03

Cheap ground, costly power



On an African tower site the ground is the smallest bill. Electricity is nearly half of IHS's cost of sales, and a plot with reliable power is worth more to a tower owner than one without.

Helios Towers paid US$46.2m under IFRS 16 leases in 2025, mainly for land and rooftop sites. That is about US$3,100 per site and 5.4% of revenue. Its power costs were US$185.5m, about US$12,600 per site and 21.7% of revenue. IHS shows the same ratio: lease cash of about US$3,500 per tower a year in the first half of 2026, annualised, against power of about US$14,300 per tower in its continuing operations. Power generation was 47.8% of IHS's cost of sales in 2025, up from 46.8% in 2024.

The reason is the grid. Supply across Helios's markets averaged 16 hours a day in 2022 and 18 hours in 2025, so sites need generators, batteries or solar to stay on air. Helios ran 56% of its sites as hybrids and 10% on solar in 2025, against 31% on hybrid or solar power in 2021, and cut diesel use in Tanzania by more than four million litres a year.

Tower owners pass much of the cost on through indexation. IHS reported that 61% of its second-quarter 2026 revenue was linked to hard currencies and power indexation. Higher global power prices raised its power costs by US$38.7m year on year in that quarter, recovered through indexation with about a one-quarter lag.

For a landowner or a land agency, power is what a plot can offer beyond space: a grid connection, the right to install solar panels, room for a shared battery yard. Each reduces the larger bill, and a tower owner may value it above a marginal rise in rent.

> *Power costs about four times as much as the ground under an African tower.*

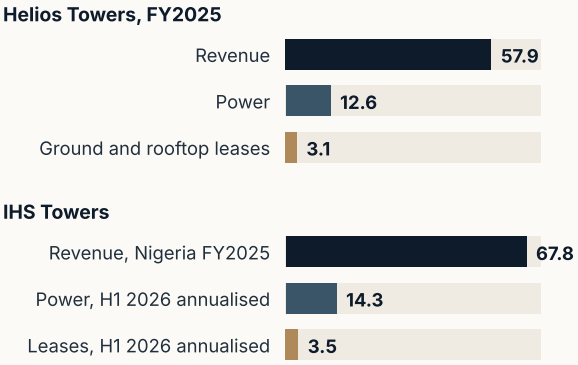
COST AND CURRENCY

Ground rent is about 5% of a tower company's revenue. In Nigeria, the currency it is paid in has moved far more.

EXHIBIT 9

Per site, power costs about four times the ground rent

US$ thousand per site a year

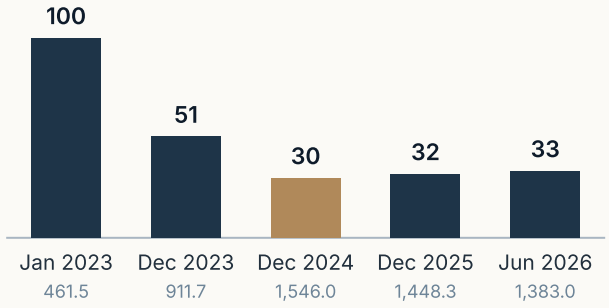


Source: Helios Towers FY2025 results (power US$185.5m; IFRS 16 lease payments US$46.2m; 14,746 sites); IHS Holding FY2025 release and 2Q26 financial statements (power US$205.4m and lease cash US$66.0m in H1 2026). **Heirstone analysis.** IHS leases are divided by 37,672 group towers including Latin America, power by 28,637 continuing towers.

EXHIBIT 10

A rent fixed in naira at the start of 2023 lost 70% of its dollar value in two years

US$ value of a fixed naira payment, index 1 January 2023 = 100; naira per US$ below



Source: IHS Holding Form 20-F FY2025 (rates at 1 Jan 2023, 31 Dec 2023 and 31 Dec 2024); IHS 4Q25 and 2Q26 releases (31 Dec 2025, 30 Jun 2026). **Heirstone analysis.**

The naira moved from 461.5 to the US dollar on 1 January 2023 to 1,546.0 on 31 December 2024, on the rates IHS uses in its accounts. A payment fixed in naira lost about 70% of its dollar value over those two years. The naira then recovered 6.7% in 2025 and closed at 1,383.0 at the end of June 2026.

Tower owners felt the other side of the move. IHS's finance costs reached US$2,042.2m in 2024, driven by foreign exchange losses on its financing, before falling to US$349.7m in 2025 as the naira steadied. The share of its revenue linked to the US dollar or euro fell from 55% in 2023 to 47% in 2025. Helios booked foreign exchange losses of US$21.8m in the first half of 2026, mainly on the Tanzanian shilling and the Ghanaian cedi, although 69% of its revenue was in hard currency.

Who carries the currency risk on the ground lease depends on its terms. A naira rent with no indexation passes the whole devaluation to the landowner, who then has every reason to reopen the price at renewal. We found no public data on how African ground leases are indexed. Escalation and currency terms are among the first questions a buyer of a tower portfolio, or a landowner signing a new lease, should ask.

70%

fall in the US$ value of a fixed naira payment, January 2023 to December 2024

US$2.04bn

IHS finance costs in 2024, driven by naira losses on financing; US$349.7m in 2025

47%

IHS revenue linked to the US$ or euro in 2025, down from 55% in 2023

THE BENCHMARK

The Heirstone Site Economics Benchmark

We compiled per-site economics for ten African tower portfolios and four comparators from company filings and deal disclosures: revenue, margin, power, ground rent, price and land tenure. Highlighted columns are the two costs of occupying a site. Blank cells were not disclosed.

EXHIBIT 11

The Heirstone Site Economics Benchmark: fourteen tower portfolios, one table

US$ thousand per site (revenue, power and ground rent per year; price once); tenancy in tenants per site

PORTFOLIO	BASIS	SITES	TENANCY	REVENUE PER SITE	EBITDA MARGIN	POWER PER SITE	GROUND RENT PER SITE	PRICE PER SITE	MULTIPLE	LAND TENURE
OPERATING PORTFOLIOS										
Helios Towers (8 African markets, Oman)	FY2025	14,746	2.17x	57.9	55%	12.6	3.1	–	–	n/d
IHS Nigeria	FY2025	15,873	1.46x	67.8	64%	–	–	–	–	See group
IHS Sub-Saharan Africa	FY2025	12,789	–	39.5	59%	–	–	–	–	See group
IHS continuing operations	H1 2026 ann.	28,637	–	–	–	14.3	3.5	–	–	Group: 89% leased; 11.4 years left
PORTFOLIOS BOUGHT OR SOLD (YEAR-1 GUIDANCE OR DEAL DATE)										
Eaton Towers (American Tower)	2019	5,700	–	45.6	64% gross	–	–	325	7.1x revenue	n/d
MTN South Africa (IHS)	2022	5,701	–	33.7	44%	–	–	72.5	2.15x revenue	n/d
Airtel Malawi (Helios)	2021	735	1.4x	31.3	–	–	–	–	–	n/d
Airtel Gabon (Helios)	2021	459	1.0x	47.9	–	–	–	–	–	n/d
Airtel Tanzania (SBA)	2021	c.1,400	–	–	–	–	–	125	–	n/d
IHS Rwanda (Paradigm)	2025	1,467	2.07x	–	–	–	–	187	8.3x EBITDA after leases	n/d
COMPARATORS OUTSIDE AFRICA										
Omantel (Helios), Gulf	2021	2,890	1.2x	20.4	68%	–	–	–	–	n/d
Zain Kuwait (IHS), Gulf	2017-20	1,499	–	–	–	–	–	–	–	n/d
IHS Latin America (Macquarie)	2026	c.9,000	–	–	–	–	–	106	–	n/d
SBA Communications (US, international)	End-2025	46,328	–	–	–	–	–	–	–	71% owned or held 20+ years; 35 years left

Source: Heirstone Site Economics Benchmark 2026, compiled from IHS Holding (20-F FY2025, 4Q25, 2Q26), Helios Towers (Annual Reports 2021, 2025; FY2025 and H1 2026 results), American Tower (2019-20 releases), SBA Communications (8-K 2021; 10-K FY2025), Capacity (Jun 2022), Developing Telecoms (Oct 2025), IHS 20-F FY2023 (Kuwait sites). **Heirstone analysis.** All per-site values divide reported totals by period-end or deal site counts. Eaton margin is gross margin. MTN South Africa revenue includes power services on about 13,000 MTN sites. IHS Sub-Saharan Africa revenue includes Rwanda to October 2025, so its per-site figure is overstated. SBA sites are 17,394 US plus 28,934 international.

Revenue per site varies threefold

From about US$20,400 a year for the Omantel portfolio to US$67,800 for IHS Nigeria. African sites earn more than the Gulf comparator, and some African revenue includes power sold with the space.

Rent is the smallest line

Ground and rooftop rent of US$3,100 to US$3,500 a year is 5.4% of revenue at Helios, and about a quarter of the power bill at both Helios and IHS.

Prices vary more than fourfold

From US$72,500 to US$325,000 a site, and from 2.15 to 7.1 times first-year revenue, with tenancy, contract currency and bundled power behind the spread.

Tenure is the least disclosed

Only IHS and SBA publish lease life and land control. A buyer should ask for the lease-expiry profile before the tenancy schedule.

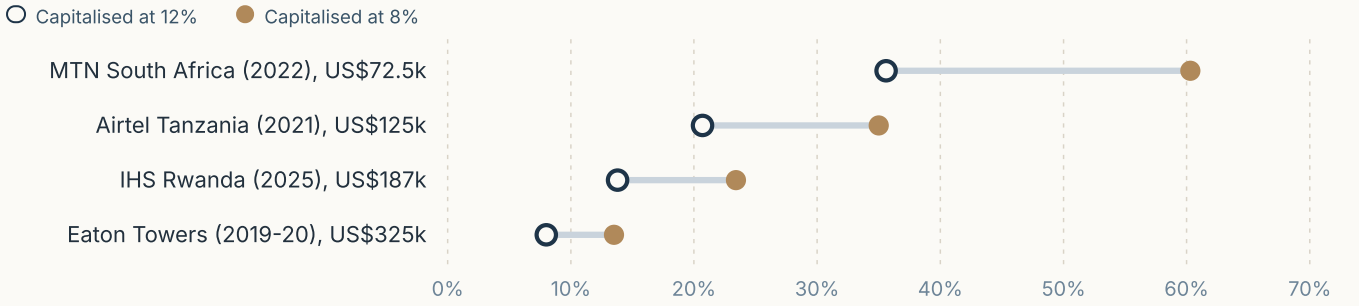
THE LANDLORD'S SHARE

At Eaton's price, the ground rent could account for less than a seventh of a site's value. The lease term underpins all of it.

EXHIBIT 12

The higher the price per site, the smaller the share of it that the ground rent could account for

Ground rent of US$3,100 to US$3,500 a year capitalised at 12% (ring) and 8% (dot), as % of the price paid per site



Source: Rents from Exhibit 9; prices from Exhibit 5. **Heirstone analysis.** Capitalisation rates of 8% and 12% are a Heirstone assumption for illustration, not observed yields; the rent range is a two-company average and will differ by site.

Exhibit 12 asks how much of each price per site the ground rent could account for if it were sold as an investment in its own right. Capitalised at 8 to 12%, a rent of US$3,100 to US$3,500 a year is worth about US$26,000 to US$44,000. That is 36 to 60% of what IHS paid per tower for MTN South Africa, but only 14 to 23% of the Rwanda price and 8 to 13% of Eaton's.

Two conclusions follow. As tenancy and contract quality rise, the land's share of a tower's value falls, so a landowner asking for a share of that value through higher rent is pushing on a small number. And the buyer's main exposure to the landowner is time: the price assumes the tower keeps earning, which it can only do while the lease runs.

A landowner therefore has more to offer than a higher rent: a longer term with renewal options priced in advance, indexation that protects both sides from the currency, the right for the tower owner to add tenants and equipment, and power. A tower owner should be willing to pay for each, because each raises what the site is worth to it and to its lenders.

The same logic applies to state land. Government land agencies, utilities and municipalities that offer standard long leases for tower sites, with grid connections, would be offering many sites on the same long terms, which no individual landowner can do.

US$26k to 44k

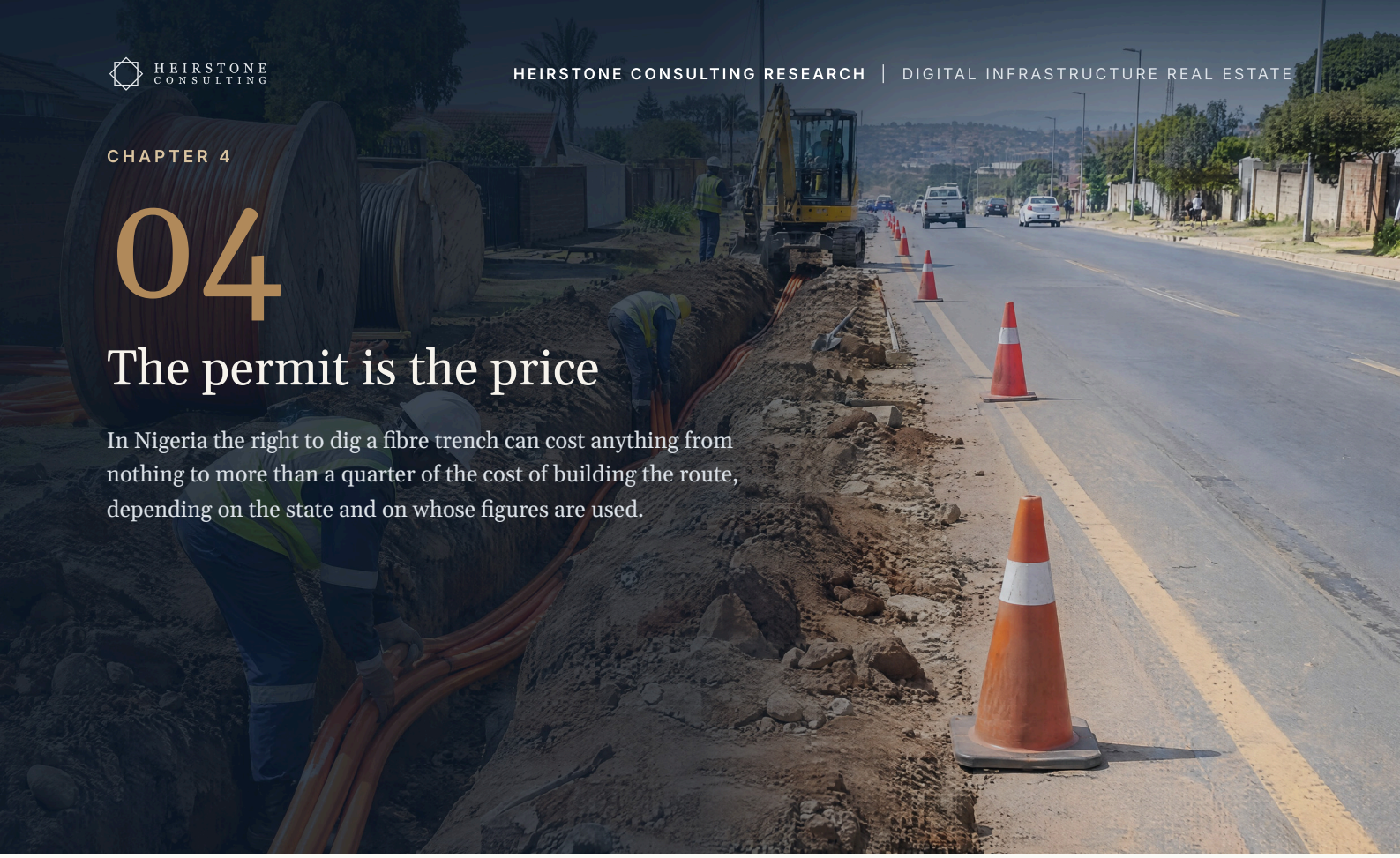
value of a typical ground rent capitalised at 8 to 12% (Heirstone illustration)

8 to 13%

share of Eaton's price per site that the ground rent could account for

36 to 60%

share of MTN South Africa's price per tower



CHAPTER 4

04

The permit is the price

In Nigeria the right to dig a fibre trench can cost anything from nothing to more than a quarter of the cost of building the route, depending on the state and on whose figures are used.

Nigeria is planning a national fibre backbone. Project BRIDGE targets 90,000 km of fibre towards a national goal of 120,000 km, at an estimated cost of US$2bn, of which US$845m has been secured: US$500m from the World Bank, US$200m from the African Development Bank and US$100m from the EBRD. Delivery is through a special-purpose vehicle in which the government holds between 25% and 49%, with 30,000 km planned in the first of five years. The average budget is about US$22,200 per route-km.

Every kilometre needs a right of way. In 2013 the National Economic Council agreed a harmonised charge of N145 per linear metre. In 2025 state charges compiled by TechCabal ran from nothing, or next to nothing, in twelve states to N9,477 per metre in Ogun, 65 times the benchmark. At NGN 1,550 per US$ (30 June 2025), Ogun's one-off charge is about US$6,114 per km, equal to about 28% of BRIDGE's average build budget. Lagos, at N6,264 per metre, is about 18%. The federal benchmark is under 0.5%.

The figures are contested. TechCabal, citing the operators' association ALTON, reported in September 2025 that only four of the states that had pledged a zero charge were applying it, and gave lower rates for some states: N850 per metre for Lagos and over N2,500 for Ogun, which would put the two at about 2.5% and at least 7% of the BRIDGE budget per km. It also reported that costs related to right of way can exceed 60% of fibre civil works costs.

120,000 km

Nigeria's national fibre target; Project BRIDGE covers 90,000 km

65x

Ogun State's charge per metre against the federal benchmark

42%

share of BRIDGE's US$2bn budget secured from the World Bank, AfDB and EBRD

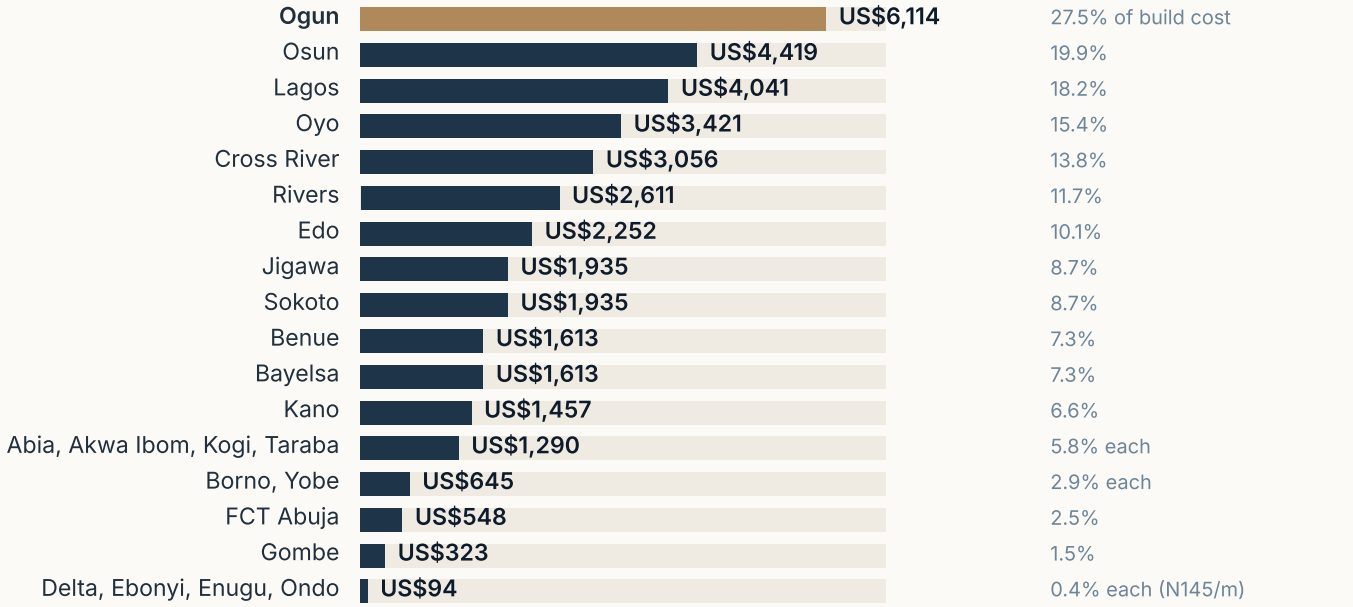
NIGERIA

Published state charges run from zero to 65 times the federal benchmark. Operators say fewer states charge zero than have pledged to.

EXHIBIT 13

Published right-of-way charges for fibre range from zero to about US$6,100 per kilometre between Nigerian states

One-off charge per route-km, US$ at NGN 1,550 per US$ (30 June 2025), and as % of Project BRIDGE average build budget (US$22,222 per km); states charging zero, and Kwara (N1 per km), not shown



Source: TechCabal, state right-of-way table (26 May 2025), compiled from state rates; Federal Ministry of Communications, Innovation and Digital Economy (Project BRIDGE); US Treasury reporting rate. **Heirstone analysis.** Rates are indicative and dated: TechCabal (12 Sep 2025, citing ALTON) gives lower figures for some states: N850 per metre for Lagos (US$548 per km, 2.5% of build cost), over N2,500 for Ogun (at least US$1,613, 7.3%) and N3,500 for Osun. Some states also levy annual fees, not shown.

TechCabal counted twelve states with free right of way in 2025. In its May 2025 table eleven charged nothing (Kaduna, Katsina, Anambra, Bauchi, Nasarawa, Adamawa, Imo, Zamfara, Kebbi, Ekiti and Plateau) and Kwara charged N1 per kilometre. In April 2025 Niger State replaced its per-metre fees with a one-time permit of N500,000, about US$323. The four highest charges in that table are all in the south-west, around Lagos.

The N145 benchmark was worth about US$190 per km at the June 2023 exchange rate and about US$94 per km at the June 2025 rate; at the end of September 2026 it was about US$109. A charge set per metre in naira and paid once falls in real terms with every devaluation, which gives states a reason to set their own, higher, rates.

A one-off charge also leaves the state with no continuing income from the route. TechCabal reported about 19,000 fibre cuts in Nigeria between January and early October 2025. Ducts laid with new roads, recorded and leased by the road owner, would give the state a recurring income and the operators a protected route.

12

Nigerian states with free right of way in 2025; four applying it in practice, according to ALTON

US$94

per km: the N145 per metre benchmark at June 2025 rates, down from US$190 in June 2023

N500,000

Niger State's one-time permit replacing per-metre fees (April 2025)

ACCESS REGIMES

South Africa grants the right to dig by statute. Nigeria prices it state by state. Egypt and Kenya negotiate it with corridor owners.

EXHIBIT 14

Five models for the right to dig, each with a different tenure risk

How fibre builders obtain access to land, streets and corridors, by model; Heirstone typology

MODEL	EXAMPLE	WHO GRANTS ACCESS	WHAT IT COSTS	WHAT IT MEANS FOR THE BUILDER'S TENURE
Statutory access	South Africa	National law (Electronic Communications Act, s22 and s24), upheld by the Constitutional Court in 2015 (6-4)	Wayleave fees set in each municipality's annual budget (draft standard by-laws, 2022)	Strong right, no landowner consent needed; 30 days' notice to the municipality; local fees vary
State-by-state negotiation	Nigeria	36 states and the FCT, under a federal benchmark of N145 per metre (2013)	Zero to N9,477 per metre, mostly one-off; Niger State: N500,000 one-time permit	Price set by location; benchmark eroded by devaluation; pledges not always applied
Utility corridor with a local layer	Kenya	Kenya Power leases dark fibre on its grid; counties levy their own charges	Kenya Power: KSh 812m revenue on 7,500+ km (FY2022/23); Nairobi County dispute over US$23.1m (2025, reported)	One corridor owner, but a second charging authority above it
Negotiated state corridor	Egypt	Suez Canal Authority agreement with Telecom Egypt for a 200 km crossing route	Not disclosed	One counterparty and a signed agreement for a strategic route; commercial terms not published
Regulator-enforced sharing	Morocco	ANRT, the telecoms regulator, enforcing infrastructure-sharing and unbundling	Fine on Maroc Telecom of MAD 3.3bn (2020) for abuse of dominance in unbundling	Access to existing ducts and networks secured through regulation

Source: Constitutional Court of South Africa, City of Tshwane v Link Africa [2015] ZACC 29; MyBroadband (21 Sep 2022, on CoGTA draft by-laws, Gazette 46920); TechCabal (21 Apr, 26 May and 12 Sep 2025; 25 Feb 2025); Kenya Power Annual Report 2022/23; Telecom Egypt Annual Report 2021; ANRT, press release on its unbundling decision (2020). **Heirstone analysis.** Typology and tenure assessment are Heirstone's.

South Africa has the strongest statutory right we found. In City of Tshwane v Link Africa (2015) the Constitutional Court upheld, by six votes to four, provisions of the Electronic Communications Act that let licensees enter land to build networks without the owner's consent. A licensee gives the municipality 30 days' written notice before building under streets and pays compensation on servitude principles. Draft by-laws of 2022 leave wayleave fees to each municipal budget.

In Kenya a second layer of charges sits above the utility: in February 2025, TechCabal reported, Nairobi County cut cables carried on Kenya Power infrastructure in a US$23.1m fee dispute with the utility. In Egypt, Telecom Egypt agreed a 200 km crossing route on the Suez Canal's west bank with the canal authority. In Morocco, ANRT fined Maroc Telecom MAD 3.3bn (about US$365m at June 2025 rates) in 2020 for abuse of dominance in unbundling.

For an investor, statutory access fixes the right and leaves the fee local; state-by-state pricing makes a route's cost depend on its path; a corridor agreement gives one counterparty and a single negotiation, with commercial terms agreed between the parties.

6-4

Constitutional Court vote upholding statutory access to land for network builders (2015)

30 days

written notice a South African licensee must give a municipality before building under its streets

CHAPTER 5

05 Ducts, corridors and landing beaches

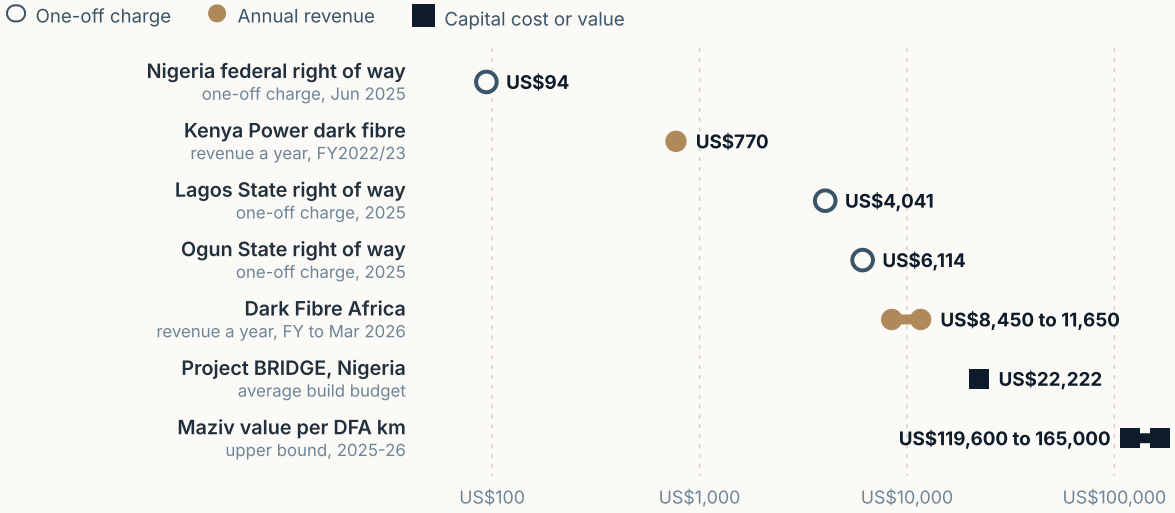
Fibre needs land in three kinds of place: city streets, long-distance corridors beside roads, railways and power lines, and the beaches where subsea cables come ashore. Revenue per kilometre on Dark Fibre Africa's mainly metropolitan network is more than ten times that on Kenya's power-grid fibre, and the coast is where capacity is most concentrated.

Kenya Power's 7,500 km of dark fibre, strung on its network across 44 of Kenya's 47 counties, earned KSh 812m (US$5.8m at June 2023 rates) from more than 25 customers in 2022/23: about US$770 per km a year.

EXHIBIT 15

A kilometre of fibre route earns about US$770 a year on Kenya's power grid and US$8,450 to US$11,650 on Dark Fibre Africa's network

US$ per route-km, log scale: one-off right-of-way charges (ring), annual revenue (gold) and capital cost or value (square)



Source: TechCabal (26 May 2025); Kenya Power Annual Report 2022/23 (KSh 812m on 7,500 km; KSh 140.55 per US$, 30 Jun 2023); MyBroadband citing CIVH and Remgro (18 Sep 2026: DFA revenue R3.001bn) and DFA website (14,500 to 20,000 km); FMCIDE Project BRIDGE; MyBroadband (25 Mar 2026: Maziv R42.491bn). Rand at ZAR 17.757 and naira at NGN 1,550 per US$ (30 Jun 2025). **Heirstone analysis.** The Maziv figure is an upper bound because the valuation includes Vumatel's home connections and Herotel.

Dark Fibre Africa, which runs metro and long-haul fibre in South Africa, earned R3.001bn in the twelve months to March 2026 and an operating profit of R1.243bn. Over a network of 14,500 to 20,000 km, depending on the source, that is R150,000 to R207,000 (US$8,450 to US$11,650) of revenue per km a year, 11 to 15 times Kenya Power's yield. Maziv's valuation equals up to about US$165,000 per DFA route-km, 7.4 times Project BRIDGE's average build budget.

Location explains much of the gap, in our view. DFA's network covers about 85% of the metropolitan areas in South Africa's main cities, according to the company, where offices, towers and homes sit close together. A power line crossing farmland is cheap to string fibre along and has few customers per kilometre.

US$770

revenue per km a year on Kenya Power's dark fibre (FY2022/23)

11 to 15x

DFA's revenue per km against Kenya Power's

CITY FIBRE AND CORRIDORS

Vodacom's Maziv deal priced homes and city ducts. Africa's corridor owners earn a fraction of that per kilometre.

EXHIBIT 16

Vodacom's entry into Maziv implies about US$813 to US$1,160 per home passed

Vodacom's investment in Maziv (Vumatel, Dark Fibre Africa and, from 2026, Herotel), completed 1 December 2025

MEASURE	RAND	US$	BASIS
Vodacom consideration for 30% of Maziv	R12.64bn	US$712m	R7.93bn cash, R4.57bn fibre assets and rights, R143m costs
Implied equity value (100%)	R42.1bn	US$2.37bn	R12.64bn / 30%; other reports give R36bn to over R47bn
Implied enterprise value	R60.1bn	US$3.39bn	Adds about R18bn of borrowings
Per home passed (2.92m homes)	R14,430 to R20,600	US$813 to US$1,160	Equity value to enterprise value
Enterprise value / operating profit	17.7x		Vumatel R2.157bn plus DFA R1.243bn, 12 months to Mar 2026
Goodwill in Vodacom's price	R6.28bn	US$354m	About 50% of the consideration
Vumatel revenue per connected home	R358 a month	US$20	R4.430bn over 1,032,226 subscribers

Source: MyBroadband (12 Jun 2026, citing Vodacom annual financial statements; 25 Mar 2026; 1 Apr 2026; 18 and 21 Sep 2026, citing CIVH and Remgro); Competition Appeal Court [2025] ZACAC 2. **Heirstone analysis.** ZAR 17.757 per US$ (US Treasury, 30 Jun 2025). Per-home values are upper bounds because DFA's network is included.

Vodacom's entry into Maziv went through two rounds of competition review. The Competition Tribunal prohibited it in October 2024; the Competition Appeal Court approved it with conditions in August 2025, including R60bn (US$3.4bn) of capital spending in South Africa over five years. Reported valuations of Maziv range from R36bn to more than R47bn, so we show the implied figures as ranges.

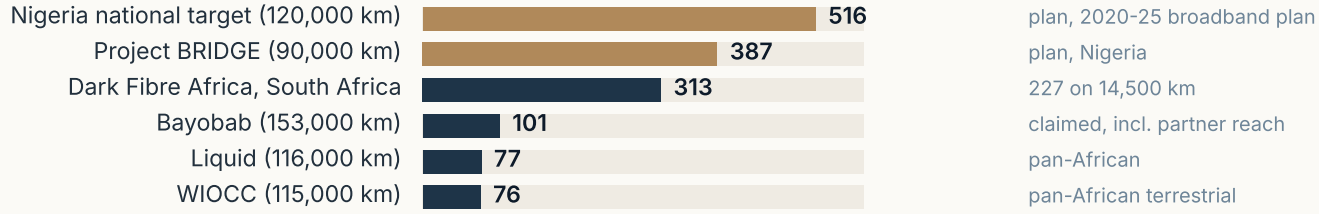
Utility and state corridors sit at the other end. Kenya Power is extending fibre to more than 53,000 public institutions under the government's Digital Superhighway. Phase3 Telecom runs aerial fibre on power lines and, with Sonatel, a 3,500 km terrestrial route from Lagos to Dakar. We found no published data on fibre leasing along Africa's railway corridors.

The pan-African networks are thin by comparison. Liquid's 116,000 km, WIOCC's 115,000 km and Bayobab's claimed 153,000 km equal 76 to 101 route-km per million Africans; in 2025 the Africa Finance Corporation and Vision Invest put US$300m into WIOCC.

EXHIBIT 17

Nigeria's fibre plan implies five times the per-capita footprint of today's largest pan-African networks

Route-km per million people; national figures use national population, pan-African networks use Africa's population (1,512 million, 2024)



Source: FMCIDE Project BRIDGE; DFA website and MyBroadband; Bayobab, Liquid Intelligent Technologies and WIOCC websites (company claims, not comparable route-km); World Bank WDI 2024. **Heirstone analysis.**

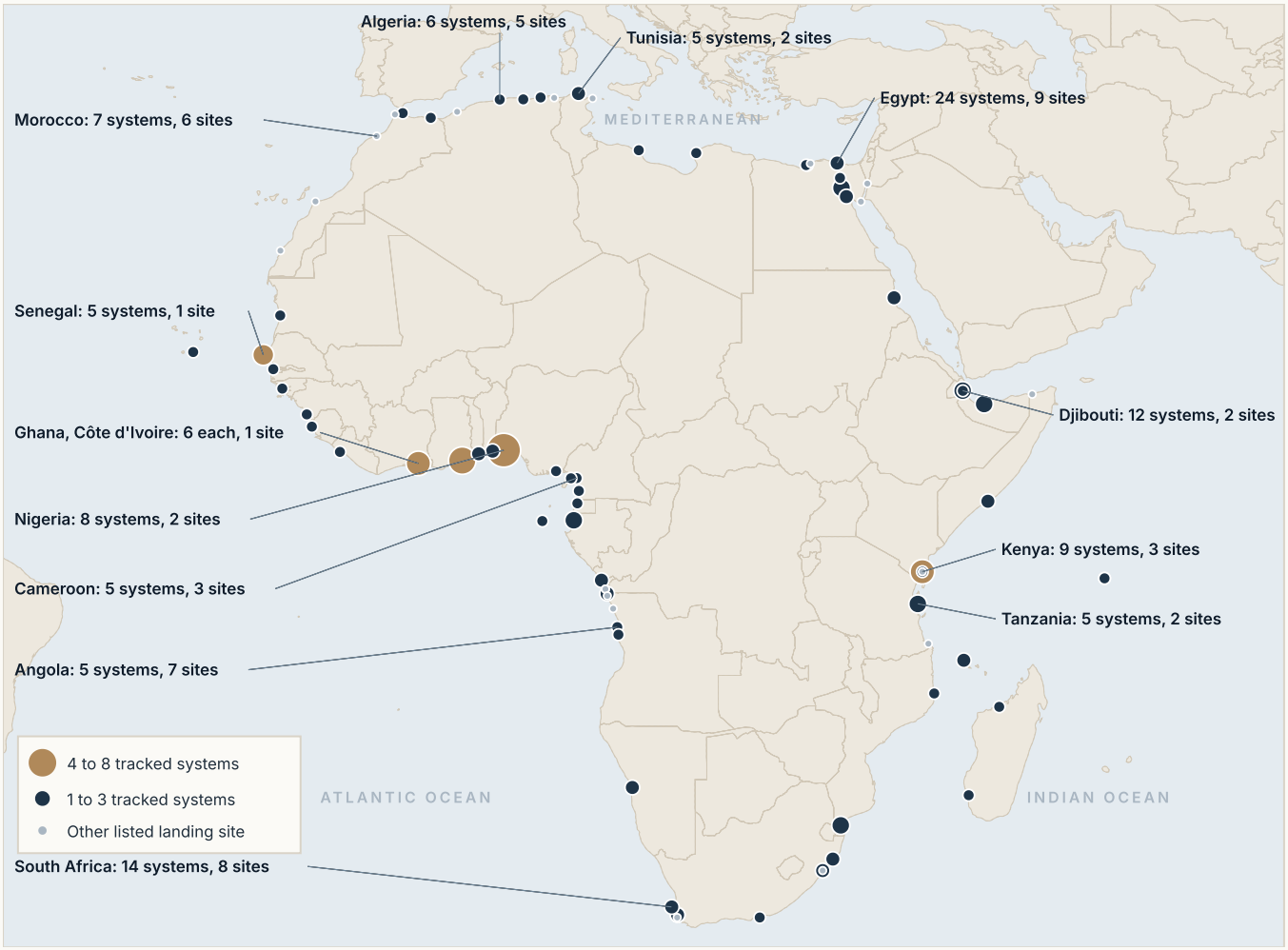
LANDING SITES

Subsea cables reach Africa through a short list of beaches, and several large countries rely on one or two of them

EXHIBIT 18

Egypt, South Africa and Angola spread their landings along the coast; Nigeria, Senegal, Ghana and Côte d'Ivoire concentrate theirs

80 cable landing sites, sized by the number of the 14 tracked systems landing there, with country totals including systems planned to 2029



Source: TeleGeography Submarine Cable Map (cable, country and landing-point pages, 2026); GeoNames gazetteer (town coordinates). **Heirstone analysis.** Positions are approximate: most are GeoNames town coordinates or TeleGeography landing points; sites without their own GeoNames entry are placed at the nearest listed town or landing point (Sangano and Muanda by Heirstone). Tracked systems: 2Africa, ACE, Africa-1, EASSy, Equiano, Glo-1, MainOne, Medusa, NCSCS, PEACE, SAT-3/WASC, SEACOM, Umoja, WACS.

24

cable systems touching Egypt, in service or planned to 2029, landing at nine sites

8

new North African landings on Medusa (8,760 km, 2026), from Morocco to Port Said

45,000

km of 2Africa, ready for service in 2024 according to TeleGeography, with four landing sites each in Egypt and South Africa

12

cable systems at Djibouti's two landing sites, the densest junction on the Red Sea after Egypt

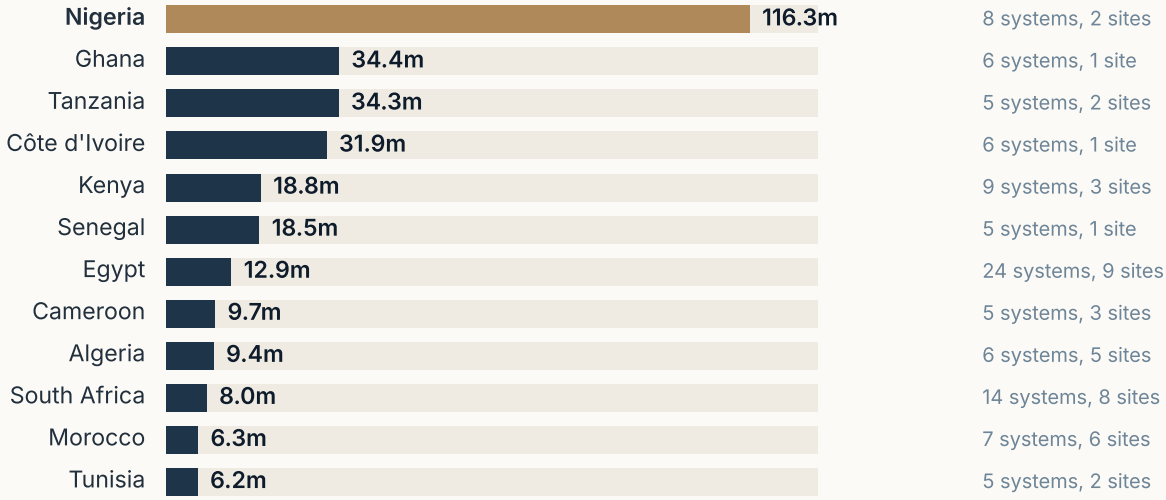
CONCENTRATION

Lagos takes all eight of Nigeria's cables. Egypt spreads twenty-four systems over nine sites.

EXHIBIT 19

Nigeria has one landing site for every 116 million people; Morocco and Tunisia one for about every 6 million

Million people per cable landing site, 2024 population; systems include those planned to 2029



Source: TeleGeography Submarine Cable Map, country pages (2026); World Bank WDI population 2024. **Heirstone analysis.**

Lagos is the landing point for eight cables, 2Africa, ACE, Equiano, Glo-1, MainOne, NCSCS, SAT-3/WASC and WACS, and Nigeria's only other site, Kwa Ibo, takes 2Africa alone. Dakar takes five systems, Accra six and Abidjan six, each at a single site. Kenya's nine systems, including Daraja (2026) and Africa-1 (2027), land at three sites on the Mombasa coast.

Egypt is the exception. Twenty-four systems touch the country at nine sites, 2Africa lands at four of them, and the Africa-1 system, whose owners include Telecom Egypt, e&, Mobily, Zain Omantel International and G42, will add landings at Ras Ghareb and Sidi Kerir in 2027. Telecom Egypt completed new landing stations at Ras Gharib, Port Said and Sidi Krir in 2021 and renovated Zafarana. Gulf operators and investors are among the owners of the next generation of African cables.

A landing site is real estate with exacting needs: a beach with protected access, a shore-end manhole, a cable route to a landing station and a corridor inland. In several West African states the state is already a co-owner of the cable: the governments of Cameroon, Equatorial Guinea, Gabon and Guinea-Bissau, and state cable companies in Liberia, Sierra Leone, The Gambia and Guinea, co-own ACE. In our view, for a coastal country with one landing site, a second landing zone with a reserved corridor inland is the most valuable piece of digital real estate a land agency can offer.

8

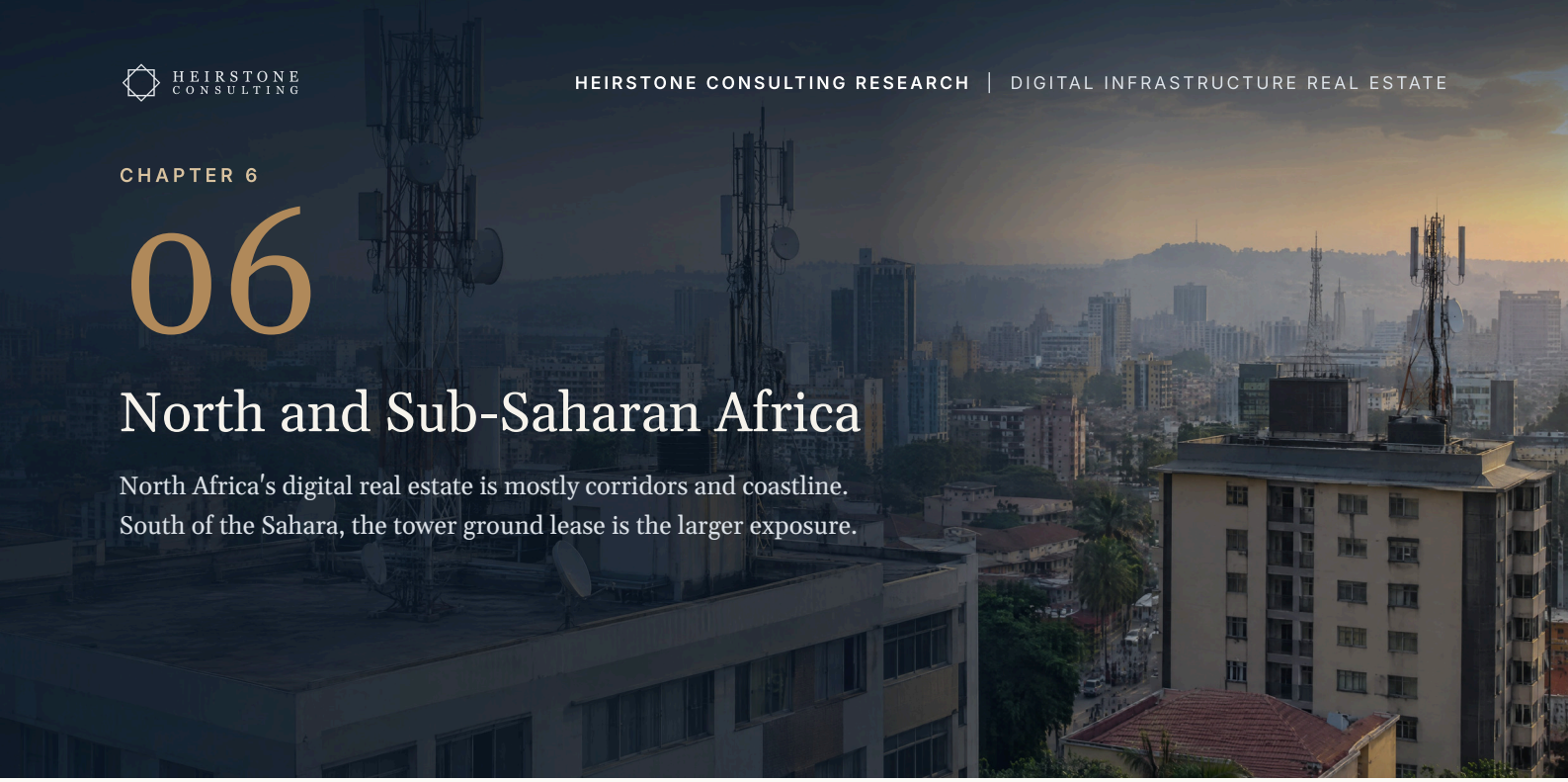
cables landing at Lagos; Nigeria's second site, Kwa Ibo, takes one

1

landing site each in Senegal, Ghana and Côte d'Ivoire

2028

planned service year for Google's Umoja, from Amanzimtoti to Western Australia



CHAPTER 6

06

North and Sub-Saharan Africa

North Africa's digital real estate is mostly corridors and coastline. South of the Sahara, the tower ground lease is the larger exposure.

NORTH AFRICA

Egypt is the land bridge for subsea cables between the Mediterranean and the Red Sea. Twenty-four cable systems touch it at nine landing sites, Telecom Egypt is a co-owner of 2Africa and Africa-1, and in September 2026 it announced a new Sharm El Sheikh to Taba subsea cable. Its 200 km crossing route on the west bank of the Suez Canal was agreed with the Suez Canal Authority, so the counterparty for that route is a single state authority. On land, Telecom Egypt's first phase under the Decent Life programme brings fibre to about 1,477 villages, with 1 million residential buildings and 14,000 government buildings, within a programme for 4,500 villages. No listed tower company yet reports an Egyptian portfolio. The 5,800-site IHS Egypt plan indicates the scale on offer if towers are separated from operators, and the terms of the land under them would then shape the valuation.

Morocco has seven cable systems across six landing sites, from Casablanca, Asilah and Tetouan to Nador, Dakhla and Tarfaya, including EllaLink to Brazil (2021), Maroc Telecom's West Africa cable (2021) and Medusa (2026). Its regulator fined Maroc Telecom in 2020 for abuse of dominance in unbundling. We found no verified public data on tower sharing or tower companies in Morocco.

Algeria, Tunisia and Libya gain from Medusa, an 8,760 km system owned by AFR-IX Telecom, which lands at Algiers and Collo, Bizerte, Benghazi and Misuratah as well as Nador, Tetouan and Port Said. Algeria has six systems across five sites and Tunisia five across Bizerte and Kelibia. Coastal plans in all three countries will need to reserve corridors from new landings to the inland network.

9

landing sites in Egypt, for 24 cable systems including planned ones: the most sites and the most systems of any country in our sample

1,477

villages in the first phase of Telecom Egypt's Decent Life fibre roll-out

MAD 3.3bn

ANRT fine on Maroc Telecom in 2020 for abuse of dominance in unbundling, about US$365m at June 2025 rates

SUB-SAHARAN AFRICA

South of the Sahara the tower lease matters most, in English- and French-speaking markets alike

ANGLOPHONE AND LUSOPHONE MARKETS

Nigeria is the largest exposure. It produced 68% of IHS's 2025 revenue from continuing operations, and IHS had 15,817 towers there at mid-2026; at the group's lease share, about 14,000 would stand on leased plots. Nigerian tenants fell 9.1% in a year as MTN agreed to vacate about 1,050 sites, while right-of-way charges across 37 jurisdictions decide the cost of each fibre route. Eight international cables land at two sites.

South Africa has the most settled rules we found: statutory access for network builders, listed-owner towers held by IHS (5,696 sites, 69.93% owned after a B-BBEE transaction), Helios (388) and American Tower, present since 2011, and a fibre group, Maziv, valued at about R42bn by Vodacom's entry. American Tower sold its South African fibre business in 2025.

Kenya has the clearest case of utility fibre, with Kenya Power's 7,500 km and county charges layered on top, and nine cable systems on the Mombasa coast. **Tanzania** combines Helios's 4,302 sites at 2.73 tenants with SBA's 1,400 towers bought from Airtel. **Ghana** has Helios's 1,099 sites at 2.44 tenants and a single landing site at Accra. **Rwanda**, with about 103 listed-owner sites per million, now has a DFI-backed owner. **Angola** has seven landing sites and a direct cable to Brazil.

FRANCOPHONE MARKETS

Côte d'Ivoire and **Cameroon** are IHS markets, with 2,669 and 2,428 towers, about 84 and 83 per million people. Abidjan takes all six of Côte d'Ivoire's cable systems at one site; Cameroon's five land at Douala, Limbe and Kribi, including SAIL to Brazil (2020), and the Republic of Cameroon is a co-owner of ACE.

Senegal has the lowest tenancy in Helios's portfolio, 1.21 tenants across 1,505 sites, five years after Helios bought 1,207 sites from Free Senegal with a commitment to build 400 more and a EUR 120m local loan. Dakar is the country's only landing site, for five systems, and the Phase3 and Sonatel terrestrial route links it to Lagos.

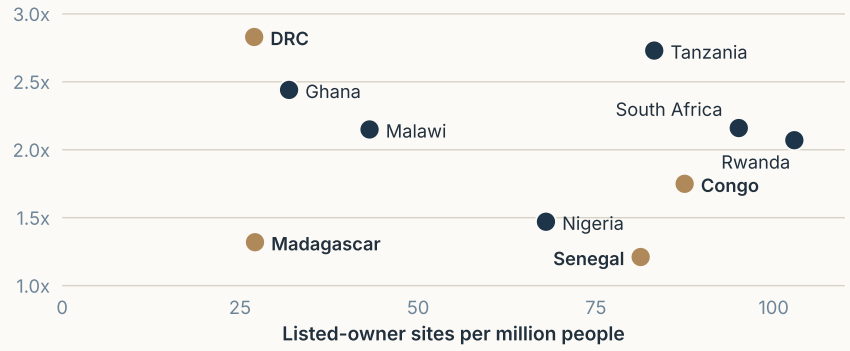
DRC has the highest tenancy in Helios's portfolio, 2.83 across 2,954 sites, and the lowest listed-owner density, 27 sites per million. 2Africa and WACS land at Muanda. **Congo-Brazzaville** (554 sites, 1.75 tenants) and **Madagascar** (867 sites, 1.32 tenants) complete Helios's francophone markets. American Tower entered **Burkina Faso** and **Niger** through Eaton in 2020, and Helios's 2021 agreement for Airtel's 459 sites in **Gabon** projected about US$47,900 of first-year revenue per site.

> *In West and Central Africa several states co-own the cable that lands on their coast.*

EXHIBIT 20

Few sites per person does not mean little sharing: DRC has the lowest density and the highest tenancy

Listed-owner sites per million people (2024 population) against tenancy, mid-2026 (Rwanda at sale, 2025); francophone markets in gold



Source: As Exhibits 4 and 6; IHS 2Q26 presentation (Nigeria). **Heirstone analysis.**

68%

of IHS revenue from continuing operations came from Nigeria in 2025

1.21x

tenancy in Senegal, the lowest in Helios's portfolio

THE COUNTRY GRID

Eighteen markets on one page

EXHIBIT 21

Country by country: towers, landings and the terms of access

Sites at 30 June 2026 unless stated; cable systems include those planned to 2029

COUNTRY	LISTED-TOWERCO SITES (OWNER)	SITES PER MILLION	TENANCY	CABLE SYSTEMS / SITES	GROUND AND ACCESS
NORTH AFRICA					
Egypt	None reported	–	–	24 / 9	Suez Canal crossing route by agreement; state-led village fibre (Decent Life)
Morocco	Not verified	–	–	7 / 6	Regulator has fined the incumbent over infrastructure sharing
Algeria	Not verified	–	–	6 / 5	Medusa lands at Algiers and Collo (2026)
Tunisia	Not verified	–	–	5 / 2	Medusa lands at Bizerte (2026)
SUB-SAHARAN AFRICA, ANGLOPHONE AND LUSOPHONE					
Nigeria	15,817 (IHS)	68	1.47x	8 / 2	State right of way from N0 to N9,477 per metre; BRIDGE 90,000 km
South Africa	6,084 (IHS, Helios)	95	2.16x (Helios)	14 / 8	Statutory access; municipal wayleave fees
Kenya	Not covered	–	–	9 / 3	Kenya Power dark fibre; county charges
Tanzania	5,702 (Helios, SBA)	83	2.73x (Helios)	5 / 2	Helios cut diesel use by 4m+ litres a year
Ghana	1,099 (Helios)	32	2.44x	6 / 1	All capacity lands at Accra
Rwanda	1,467 (Paradigm)	103	2.07x	Landlocked	DFI-backed owner since 2025
Zambia	2,027 (IHS)	95	–	Landlocked	–
Angola	Not covered	–	–	5 / 7	Direct cable to Brazil (SACS, 2018)
SUB-SAHARAN AFRICA, FRANCOPHONE					
Côte d'Ivoire	2,669 (IHS)	84	–	6 / 1	All capacity lands at Abidjan
Senegal	1,505 (Helios)	81	1.21x	5 / 1	All capacity lands at Dakar; Lagos-Dakar terrestrial route
Cameroon	2,428 (IHS)	83	–	5 / 3	State co-owns ACE; cable to Brazil (SAIL, 2020)
DRC	2,954 (Helios)	27	2.83x	Muanda	Highest Helios tenancy, lowest density
Congo-Brazzaville	554 (Helios)	88	1.75x	Pointe-Noire	–
Madagascar	867 (Helios)	27	1.32x	2 sites	Mahajanga (2Africa), Toliara (EASSy)

Source: As Exhibits 3, 4, 14, 18 and 19; Helios Towers Annual Report 2025; Telecom Egypt Annual Report 2021; Phase3 Telecom; TeleGeography (Angola, Cameroon, Madagascar landings). Not covered: no listed-owner country data in our sources. **Heirstone analysis.**

North Africa has well-connected coastlines and no listed tower companies, so its questions are about corridors, landing zones and, if towers are separated from operators, the land under them. Nigeria combines the largest tower exposure with the most fragmented access rules and the most concentrated landing. Francophone West and Central Africa pairs low tenancy in some markets with single landing sites at Abidjan and Dakar.

The Gulf appears as a comparator and as capital. Helios's Omani towers, at 505 per million people and about US$20,400 of first-year revenue per site, show a far denser market earning less per site. Gulf groups co-own the new cables: center3 is an owner of 2Africa, and Africa-1's owners include e&, Mobily, Zain Omantel International and G42 alongside Telecom Egypt.

18

markets compared on towers, landings and access

505

Helios sites per million people in Oman, the Gulf comparator

CHAPTER 7

07 The landlord test

In our view six questions decide whether a land right under a tower or a fibre route is bankable. We apply them to five rights described in this report.

Each question comes from the evidence in earlier chapters: term from the IHS and SBA leases, currency from the naira, counterparties from the 25,500 leases and 37 Nigerian charging authorities, consent from South Africa's statute, co-use from tenancy, and power from the cost stack. Any landowner, land agency or investor can answer them before signing.

EXHIBIT 22

No land right we examined passes all six questions; the most valuable one fails three

The Heirstone Landlord Test applied to five land rights under African towers and fibre

● Passes ◑ Partly ○ Fails – Not published

QUESTION	TOWER GROUND LEASES (IHS, AFRICA)	NIGERIAN STATE RIGHT OF WAY	SOUTH AFRICAN STATUTORY ACCESS	KENYA POWER UTILITY CORRIDOR	SUEZ CANAL CROSSING ROUTE
1 Term Does the right outlast the investor's hold? Mature-market mark: SBA, 35 years left	○ 11.4 years left; a third end by 2030	– Term not published	● Statutory right	– Customer terms not published	– Not disclosed
2 Currency Is the payment indexed, or linked to a hard currency?	– Indexation not disclosed	○ N145 benchmark halved in US$	◑ Set in each municipal budget	– Not published	– Not disclosed
3 Counterparties How many parties must be agreed with to secure the asset?	○ About 25,500 landowners	○ 37 charging authorities	◑ Each municipality	◑ One utility, plus county charges	● One authority, 200 km
4 Consent Is access granted by statute or standard terms on a fixed timetable?	◑ Renewed lease by lease	○ Zero pledges applied in 4 states	● 30 days' notice; no owner consent	◑ County charges disputed	● Agreement signed
5 Co-use Can the right carry more tenants or cables without a new negotiation?	● 1.2x to 2.8x tenancy; amendments	– Not published	● By-laws favour existing facilities	● 25+ dark-fibre customers	– Not disclosed
6 Power Does the site come with reliable energy?	○ Grid 16-18 hours; power 4x rent	Not applicable	Not applicable	● On the grid itself	Not applicable

Source: Evidence as cited in Exhibits 2, 6, 9, 13 and 14: IHS 20-F FY2025 and 2Q26 presentation; Helios Annual Report 2025; TechCabal (2025); Constitutional Court [2015] ZACC 29; MyBroadband (2022); Kenya Power Annual Report 2022/23 and 2024/25; Telecom Egypt Annual Report 2021. **Heirstone assessment.**

The tower ground lease, which supports the most value, fails on term, counterparties and power; its indexation is not disclosed. The Nigerian state right of way fails on currency, counterparties and consent. South Africa's statutory access passes three of the five questions that apply; Kenya Power's corridor passes two and partly passes two. The Suez Canal route passes the two questions on which its terms are public and fails none. Of the 30 cells, eight cannot be filled from public information, so an investor pricing these rights has to assume the missing terms.

The test also shows where the value lies for those who grant land rights. A landowner who offers a 20-year term, indexation, the right to add tenants and a grid connection answers four of the six questions in the tower owner's favour. An authority that publishes one tariff, grants access on a timetable and keeps a register of what lies in its corridors should make every route in its territory cheaper to finance.

8

cells that pass, out of 30

6

cells that fail, all on tower ground leases and Nigerian state rights of way

WHAT TO DO

Lengthen, index and publish the land rights, then price them

Towers, fibre and the operator contracts behind them are already financed on infrastructure terms. Each audience below can bring the land rights beneath them closer to those terms.

FOR

Governments, regulators and land agencies

Replace one-off per-metre charges with a single published national tariff, indexed and collected through one window. Nigeria's N145 benchmark lost half its dollar value in two years; on TechCabal's figures Ogun's charge equalled 28% of build cost.

Reserve a second cable landing zone, with a protected corridor inland, wherever a country depends on one or two sites: Nigeria, Senegal, Ghana, Côte d'Ivoire.

Offer state land for tower sites on standard 20-year leases with grid connections, especially where towers are being sold or separated from operators.

Publish the terms not yet public: lease terms on state land, corridor tariffs and landing-site agreements. Eight of the 30 cells in our test could not be filled.

FOR

Road, rail and power authorities

Lay ducts with new roads and register them. Lease them per duct and per km on the urban approaches, where Dark Fibre Africa earns 11 to 15 times more per km than utility fibre.

Price long-haul corridor access as volume business with published standard terms, and settle county or municipal charges before a dispute arises. Kenya Power's 7,500 km earn about US$770 per km a year.

Treat canal, rail and pipeline corridors as fibre real estate. Telecom Egypt's 200 km Suez Canal route shows a single authority can grant a strategic right of way.

FOR

Landowners and investors in ground leases

Trade term for price: to a tower owner, a 20-year lease with priced renewals and indexation is likely to be worth more than a higher rent on a short lease: rent is about 5% of revenue.

Sell the plot with power: a grid connection or room for solar addresses a cost four times the rent.

For pension funds and land funds: ground leases under towers, bought, lengthened and aggregated, would be a local-currency asset with tower companies and operators as tenants. IHS alone implies about 25,500 of them.

FOR

Operators, tower companies and lenders

Lock tenure before a sale or refinancing: a third of IHS's leases expire by the end of 2030 and 14% within two years.

Report lease expiry, indexation and land control alongside tenancy. Only IHS and SBA among the companies we reviewed disclose them.

In fibre, underwrite homes passed and city ducts, using Maziv's US$813 to US$1,160 per home and 17.7 times operating profit as the reference, and discount long-haul routes for access risk by state.

How Heirstone works on digital infrastructure land

We help governments, land agencies, landowners, operators and investors decide what digital real estate is worth and on what terms to hold it: site and corridor valuation, ground-lease and wayleave structuring, highest-and-best-use studies for state land and corridors, feasibility modelling (DCF, IRR, NPV and Monte Carlo) and benchmarking of the kind in this report.

Get in touch

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NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

Site: a tower or rooftop location. **Tenancy ratio:** tenants (or tenancies) divided by sites; IHS calls it the colocation rate. **Lease amendment:** added equipment by an existing tenant. **Ground lease:** a lease of the land under a tower. **Right of way or wayleave:** the right to place cables in, under or over land or streets owned by another. **Route-km:** length of a fibre route regardless of the number of cables or fibres in it. **Homes passed:** homes a fibre network can connect without new street works. **Build-to-suit:** a new tower built for an operator. **EV:** enterprise value. **DFI:** development finance institution.

Method

Figures are as published by the sources listed, for the period stated. Calculations are labelled Heirstone analysis and use only the inputs shown with them. Per-site and per-km values divide reported totals by period-end or deal counts. Exchange rates: US Treasury reporting rates at 30 June 2023 (NGN 765, KES 140.55), 30 June 2025 (NGN 1,550, ZAR 17.757, MAD 9.039) and 30 September 2026 (NGN 1,326.78); the naira series in Exhibit 10 uses IHS's reporting rates. The naira has been devalued over the periods compared, so dollar values from different years are not like for like. Capitalisation and discount rates of 8 to 12% are illustrative assumptions. Where sources conflict (Maziv valuation, DFA route-km, Nigerian state charges) we show the range. Single-source figures are attributed in the text. Company route-km claims are not comparable with each other. Map positions are approximate. The Landlord Test grades are Heirstone's judgement on the evidence cited.

Use of this report

This report is published for general information. It is not investment, legal or valuation advice and should not be relied on as such. It uses public information only. Images are illustrative.

SUMMARY AND CHAPTER 1

IHS Holding Ltd, Form 20-F FY2025 (Mar 2026), FY2021 and FY2023; 4Q25 (16 Mar 2026), 1Q26 (12 May 2026) and 2Q26 (11 Aug 2026) releases, presentations and financial statements; SBA Communications, Form 10-K FY2025 and 8-K (2 Jun 2021); Helios Towers, H1 2026 results (30 Jul 2026), FY2025 results (12 Mar 2026), Annual Reports 2020, 2021, 2022 and 2025; American Tower, Form 10-K FY2011, FY2019 and FY2025, press releases (30 May 2019, 2 Jan 2020); Africa Mobile Networks website; CTIA; World Bank WDI (SP.POP.TOTL, 2024).

CHAPTERS 2 AND 3

As above; Capacity, formerly TowerXchange (Jun 2022); Developing Telecoms (Oct 2025); IHS Towers, Our history (2026); IHS 1Q26 earnings release (MTN offer).

CHAPTER 4

Federal Ministry of Communications, Innovation and Digital Economy, Project BRIDGE; TechCabal (19 Apr 2024; 25 Feb, 21 Apr, 26 May, 12 Sep and 8 Oct 2025); Constitutional Court of South Africa, City of Tshwane v Link Africa [2015] ZACC 29; MyBroadband (21 Sep 2022); Telecom Egypt Annual Report 2021; ANRT, unbundling decision and press release (2020); US Treasury Fiscal Data, reporting rates of exchange.

CHAPTER 5

Kenya Power Annual Report 2022/23 and Abridged Annual Report 2024/25; Competition Appeal Court of South Africa [2025] ZACAC 2; MyBroadband (25 Mar, 1 Apr, 12 Jun, 18 Sep and 21 Sep 2026); Dark Fibre Africa, Liquid Intelligent Technologies, WIOCC, Bayobab, CSquared and Phase3 Telecom websites; TeleGeography Submarine Cable Map (cable, country and landing-point pages, 2026); GeoNames gazetteer (map positions).

CHAPTERS 6 AND 7

As above; Telecom Egypt investor relations (2026); Helios Towers Annual Report 2021 (Senegal facility; Gabon).

CONFLICTING AND SINGLE-SOURCE FIGURES

MTN South Africa tower count: 5,701 at announcement, 5,691 in IHS's 20-F. Maziv valuation: R36bn, R42.1bn, R42.5bn and over R47bn in different reports; Vodacom's consideration reported as R12.64bn (annual financial statements) and earlier as R6.1bn cash plus R4.89bn of assets. DFA route-km: 14,500 (MyBroadband) against over 20,000 (company). Nigerian right of way: Lagos N6,264 per metre and Ogun N9,477 (TechCabal, May 2025) against N850 and over N2,500 (TechCabal citing ALTON, Sep 2025); TechCabal gives Kwara's charge as N1 per kilometre. Constitutional Court vote in Link Africa: six to four. Single-source figures: Africa's 170,000 towers (Africa Mobile Networks); Nigerian fibre cuts and right-of-way pledges (TechCabal).

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